Juan Camilo Lozano Carrillo, Ph.D.

Professional Summary: My multidisciplinary background spans biomedical engineering, cybersecurity, and data science. Throughout my doctoral research in cybersecurity, I applied pattern recognition, statistical modeling, and predictive analysis to complex security problems—transferable approaches that allow me to examine challenges from multiple perspectives and develop effective solutions across domains. I am passionate about using data analytics, mathematical modeling, and software development to advance technological innovation. In fact, I implemented IDCBISMATCH a genetical matching software that is currently used by a Health Institute in Bogotá (IDCBIS) to treat blood disseases in children, this is one of my most rewarding projects due to its impact on the health sector of my Country.

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https://juanclozano.com/

https://scholar.google.com/citations?user=REWcf5AAAAAJ

Fluent in English and Spanish

Education

Jan 2019 – Dec 2024	**Ph.D. in Computer Science and Engineering**, University of California, Santa Cruz.
Jan 2019 – May 2023	**M.Sc. in Computer Science and Engineering**, University of California, Santa Cruz.
Feb 2011 – Nov 2015	**M.Sc. in Biomedical Engineering**, Universidad Nacional de Colombia.
Feb 2005 – May 2010	**BSc. in Electronics Engineering**, Universidad Nacional de Colombia.

Awards and Achievements

2024	**CPSIoTSec 2024 Best Paper Award**, The Joint Workshop on CPS & IoT Security and Privacy.
2023	**Commander's Award**, 2023 Cyber Recon by United States Cyber Command.
2018	**Fulbrigth Scholarship**, US Department of State and Colombia's MinTIC.
2011	**Outstanding Graduate Student**, Universidad Nacional de Colombia.

Work Experience

Jun 2026 - Current

Technology Consultant at The District Institute of Science, Biotechnology and Innovation in Health (IDCBIS); Bogotá, Colombia.
Description: I work closely with biology scientists and physicians to develop a digital transformation and data governance for the Umbilical Cord Bank area, and the institute implementation of international standards for quality and interoperability.

Oct 2025 - Sep 2026

Data Scientist and Governance consultant. at the Land Use Planning Office of Medellín's Town Hall; Medellín, Colombia.
Description: I supported Medellin's Land Use planning by interpreting, analyzing and generating statistical models from geo-referenced data collected on the mandatory update of a Land Use Plan. I also extracted topics from huge volumes of information, I drew conclusions after running statistics on data. I developed some solutions using R, other using Pandas.

Work Experience (continued)

Mar 2024 - Dec 2024 — **Graduate Student Instructor.** Applied Discrete Mathematics course (120 students, manager of a team of 6); Computer Science Department; Baskin School of Engineering; University of California, Santa Cruz.

Sep 2019 – Dec 2024 — **Teaching Assistant.** Computer Security, Cyber Physical Systems Security (Including Medical Devices safety and security) & Applied Discrete Mathematics courses; Computer Science Department; Baskin School of Engineering; University of California, Santa Cruz.

Jan 2018 - Jul 2018 — **Software Developer** at The District Institute of Science, Biotechnology and Innovation in Health (IDCBIS); Bogotá, Colombia.
Description: I designed and implemented a matching software for the Human Leukocyte Antigen for IDCBIS' Umbilical Cord Bank using Python and proposed validation strategies, additionally I implemented a Graphic User Interface.

Aug 2017 - Dec 2017 — **Patent Reviewer** at the Colombian National Patent Office; Bogotá, Colombia
Description: I reviewed novelty claims of patent files in the tech field, delivering reports that either accepted or denied a patent issued by the Colombian government.

Research Publications

Academic

1. K. Koneru, J. Lozano, J. Castellanos, E. Zambon, and A. Cardenas, "Unveiling the operation and configuration of a real-world bulk substation network," in *EAI SmartSP 2024 - 2nd EAI International Conference on Security and Privacy in Cyber-Physical Systems and Smart Vehicles*, IEEE, 2024.
2. L. Salazar, S. Castro, J. Lozano, K. Koneru, E. Zambon, B. Huang, R. Baldick, M. Krotofil, A. Rojas, and A. Cardenas, "A tale of two industroyers: It was the season of darkness," in *2024 IEEE Symposium on Security and Privacy*, ser. 2024 IEEE Symposium on Security and Privacy, IEEE, 2024.
3. L. Salazar, E. Morales, J. Lozano, C. Medrano, and A. Cardenas, "Icsnet: A hybrid-interaction honeynet for industrial control systems," in *Proceedings of the Sixth Workshop on CPSIoT Security and Privacy (CPSIoTSec'24)*, 2024.
4. J. Lozano, K. Koneru, N. Ortiz, and A. A. Cardenas, "Digital substations and iec 61850: A primer," *IEEE Communications Magazine*, vol. 61, no. 6, 2023, ISSN: 0163-6804.
5. J. Lozano, K. Koneru, J. H. Castellanos, and A. A. Cardenas, "Timing analysis of goose in a real-world substation," in *2022 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*, ser. 2022 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), IEEE, 2022, ISBN: 1-66543-255-1.
6. D. Vanegas, C.-C. Galindo, I.-A. Páez-Gutiérrez, L.-X. González-Acero, P.-T. Medina-Valderrama, J.-C. Lozano, B. Camacho-Rodríguez, and A.-M. Perdomo-Arciniegas, "Human leukocyte antigen and red blood cells impact umbilical cord blood cd34+ cell viability after thawing," *International Journal of Molecular Sciences*, vol. 20, no. 19, 2019, ISSN: 1422-0067. DOI: 10.3390/ijms20194875.

Skills

Coding — Java, Python, R, SQL, Matlab, C++

Skills (continued)

Misc.	Research, Leadership, problem solving, abstraction and mathematical modeling, Data Analysis, Data Science, Machine Learning, Cybersecurity and Privacy.
Languages	Full proficiency: Spanish and English, Basic: Portuguese.